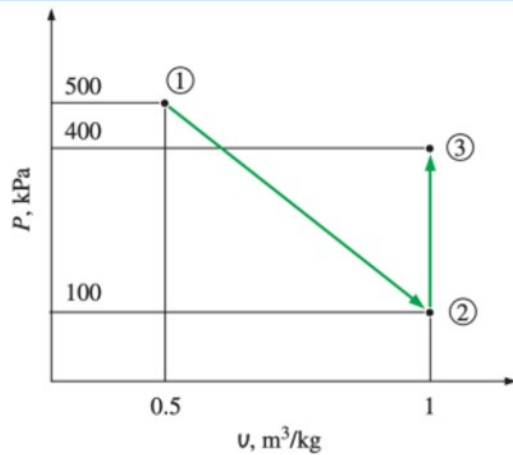


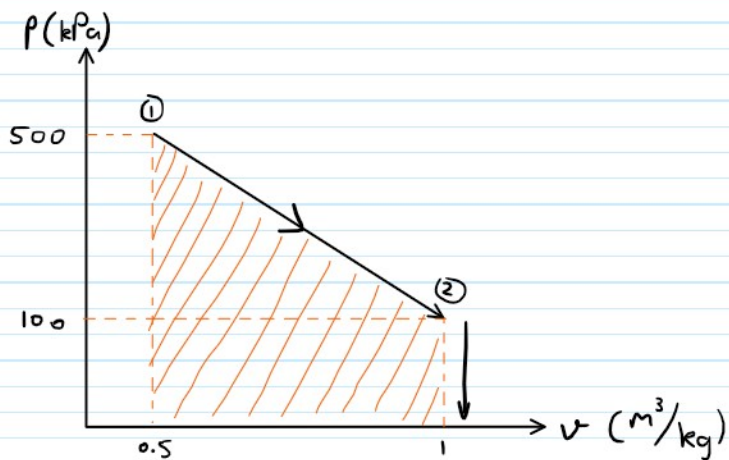
Boundary Work (Week 4, tutorial 1)

Monday, 11 August 2025 1:56 pm

- Calculate the total work, in kJ, for process 1–3 shown in the figure below when the system consists of 3 kg of nitrogen.



Process 1 to 2 (expansion)



$$W_{B(1 \rightarrow 2)} = A_{1 \rightarrow 2}$$

$$W_{B(1 \rightarrow 2)} = A_{\text{triangle}} + A_{\text{rectangle}}$$

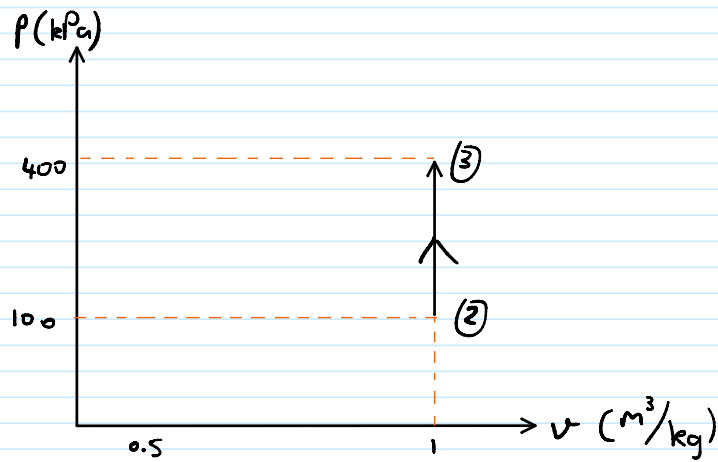
$$W_{B(1 \rightarrow 2)} = \frac{(1 - 0.5)(500 - 100)}{2} + (1 - 0.5)(100 - 0)$$

$$W_{B(1 \rightarrow 2)} = 150 \text{ J/kg} \left\{ \begin{array}{l} \text{This value is positive, which means work is being} \\ \text{done BY the gas on its surroundings between 1 and 2} \\ \text{(e.g., the gas is expanding and lifting up a piston).} \end{array} \right.$$

Process 2 to 3 (constant volume/isochoric)

$P \text{ (kPa)}$

PROCESS 2 TO 3 (CONSTANT VOLUME/ISOCORIC)



$$W_B(2 \rightarrow 3) = A_{2 \rightarrow 3}$$

$$W_B(2 \rightarrow 3) = 0 \quad \left\{ \begin{array}{l} \text{Since the volume is constant throughout process 2-3, this} \\ \text{means that the gas isn't doing any work on its} \\ \text{surroundings. This is shown in the P-v diagram with a} \\ \text{vertical line.} \end{array} \right.$$

$$W_B(1 \rightarrow 3) = W_B(1 \rightarrow 2) + W_B(2 \rightarrow 3)$$

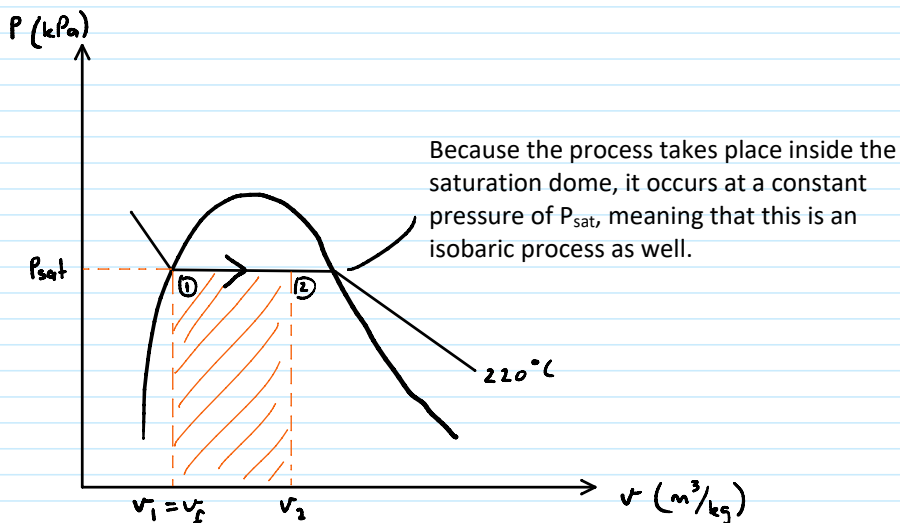
$$W_B(1 \rightarrow 3) = 150 \text{ J/kg}$$

$$\text{then, } W_B(1 \rightarrow 3) = m W_B(1 \rightarrow 3) \leftarrow \text{Converting specific work to total work}$$

$$W_B(1 \rightarrow 3) = 3 \times 150$$

$$W_B(1 \rightarrow 3) = 450 \text{ J, or } 0.45 \text{ kJ}$$

2. 2 m^3 of saturated liquid water at 220°C is expanded isothermally in a closed system until its quality is 80 percent. Determine the total work produced by this expansion, in kJ.



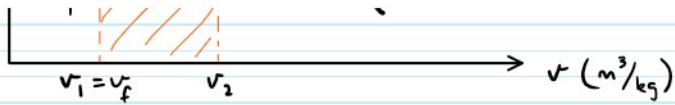


TABLE A-4

Saturated water—Temperature table (Concluded)

Temp., T °C	Sat. press., P _{sat} kPa	Specific volume, m ³ /kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg·K		
		Sat. liquid, v _f	Sat. vapor, v _g	Sat. liquid, u _f	Evap., u _{fg}	Sat. vapor, u _g	Sat. liquid, h _f	Evap., h _{fg}	Sat. vapor, h _g	Sat. liquid, s _f	Evap., s _{fg}	Sat. vapor, s _g
205	1724.3	0.001164	0.11508	872.86	1723.5	2596.4	874.87	1920.0	2794.8	2.3776	4.0154	6.3930
210	1907.7	0.001173	0.10429	895.38	1702.9	2598.3	897.61	1899.7	2797.3	2.4245	3.9318	6.3563
215	2105.9	0.001181	0.094680	918.02	1681.9	2599.9	920.50	1878.8	2799.3	2.4712	3.8489	6.3200
220	2319.6	0.001190	0.086094	940.79	1660.5	2601.3	943.55	1857.4	2801.0	2.5176	3.7664	6.2840
225	2549.7	0.001199	0.078405	963.70	1638.6	2602.3	966.76	1835.4	2802.2	2.5639	3.6844	6.2483
230	2797.1	0.001209	0.071505	986.76	1616.1	2602.9	990.14	1812.8	2802.9	2.6100	3.6028	6.2128
235	3062.6	0.001219	0.065300	1010.0	1593.2	2603.2	1013.7	1789.5	2803.2	2.6560	3.5216	6.1775
240	3347.0	0.001229	0.059707	1033.4	1569.8	2603.1	1037.5	1765.5	2803.0	2.7018	3.4405	6.1424
245	3651.2	0.001240	0.054656	1056.9	1545.7	2602.7	1061.5	1740.8	2802.2	2.7476	3.3596	6.1072
250	3976.2	0.001252	0.050085	1080.7	1521.1	2601.8	1085.7	1715.3	2801.0	2.7933	3.2788	6.0721
255	4322.9	0.001263	0.045941	1104.7	1495.8	2600.5	1110.1	1689.0	2799.1	2.8390	3.1979	6.0369
260	4692.3	0.001276	0.042175	1128.8	1469.9	2598.7	1134.8	1661.8	2796.6	2.8847	3.1169	6.0017
265	5085.3	0.001289	0.038748	1153.3	1443.2	2596.5	1159.8	1633.7	2793.5	2.9304	3.0358	5.9662
270	5503.0	0.001303	0.035622	1177.9	1415.7	2593.7	1185.1	1604.6	2789.7	2.9762	2.9542	5.9305
275	5946.4	0.001317	0.032767	1202.9	1387.4	2590.3	1210.7	1574.5	2785.2	3.0221	2.8723	5.8944
280	6416.6	0.001333	0.030153	1228.2	1358.2	2586.4	1236.7	1543.2	2779.9	3.0681	2.7898	5.8579
285	6914.6	0.001349	0.027756	1253.7	1328.1	2581.8	1263.1	1510.7	2773.7	3.1144	2.7066	5.8210
290	7441.8	0.001366	0.025554	1279.7	1296.9	2576.5	1289.8	1476.9	2766.7	3.1608	2.6225	5.7834
295	7999.0	0.001384	0.023528	1306.0	1264.5	2570.5	1317.1	1441.6	2758.7	3.2076	2.5374	5.7450
300	8587.9	0.001404	0.021659	1332.7	1230.9	2563.6	1344.8	1404.8	2749.6	3.2548	2.4511	5.7059
305	9209.4	0.001425	0.019932	1360.0	1195.9	2555.8	1373.1	1366.3	2739.4	3.3024	2.3633	5.6657
310	9865.0	0.001447	0.018333	1387.7	1159.3	2547.1	1402.0	1325.9	2727.9	3.3506	2.2737	5.6243
315	10556	0.001472	0.016849	1416.1	1121.1	2537.2	1431.6	1283.4	2715.0	3.3994	2.1821	5.5816
320	11284	0.001499	0.015470	1445.1	1080.9	2526.0	1462.0	1238.5	2700.6	3.4491	2.0881	5.5372
325	12051	0.001528	0.014183	1475.0	1038.5	2513.4	1493.4	1191.0	2684.3	3.4998	1.9911	5.4908
330	12858	0.001560	0.012979	1505.7	993.5	2499.2	1525.8	1140.3	2666.0	3.5516	1.8906	5.4422
335	13707	0.001597	0.011848	1537.5	945.5	2483.0	1559.4	1086.0	2645.4	3.6050	1.7857	5.3907
340	14601	0.001638	0.010783	1570.7	893.8	2464.5	1594.6	1027.4	2622.0	3.6602	1.6756	5.3358
345	15541	0.001685	0.009772	1605.5	837.7	2443.2	1631.7	963.4	2595.1	3.7179	1.5585	5.2765
350	16529	0.001741	0.008806	1642.4	775.9	2418.3	1671.2	892.7	2563.9	3.7788	1.4326	5.2114
355	17570	0.001808	0.007872	1682.2	706.4	2388.6	1714.0	812.9	2526.9	3.8442	1.2942	5.1384
360	18666	0.001895	0.006950	1726.2	625.7	2351.9	1761.5	720.1	2481.6	3.9165	1.1373	5.0537
365	19822	0.002015	0.006009	1777.2	526.4	2303.6	1817.2	605.5	2422.7	4.0004	0.9489	4.9493
370	21044	0.002217	0.004953	1844.5	385.6	2230.1	1891.2	443.1	2334.3	4.1119	0.6890	4.8009
373.95	22064	0.003106	0.003106	2015.7	0	2015.7	2084.3	0	2084.3	4.4070	0	4.4070

Initial state

- Saturated liquid
- 220 °C
- 2 m³

From A-4:

$$v_f @ 220^\circ\text{C} = 0.00119 \text{ m}^3/\text{kg}$$

$$m = \frac{V_1}{v_1} \quad \left. \begin{array}{l} \text{We can use the system's initial volume, and initial} \\ \text{specific volume to find the mass of water present} \end{array} \right\}$$

$$m = \frac{2}{0.00119}$$

$$m = 1680.6723 \text{ kg} \quad \text{— Mass of water in system. Since the system is closed, this is constant}$$

Final state

- Saturated mixture
- X = 0.8 (80% has converted to vapour)
- 220 °C

$$X = \frac{v_2 - v_f}{v_g - v_f} \quad \left\{ \begin{array}{l} \text{The second state is a mixture (both phases are} \\ \text{present), and so the quality (vapour fraction) must be} \\ \text{used to find the water's exact properties} \end{array} \right.$$

$$X = \frac{v_2 - v_f}{v_g - v_f} \left\{ \begin{array}{l} \text{The second state is a mixture (both phases are} \\ \text{present), and so the quality (vapour fraction) must be} \\ \text{used to find the water's exact properties} \end{array} \right.$$

$$v_2 = v_f + X(v_g - v_f)$$

From A-4:

$$v_2 = 0.00119 + 0.8(0.086094 - 0.00119)$$

$$v_2 = 0.0691132 \text{ m}^3/\text{kg}$$

Work done

- Because the process occurs within the saturation region, this means that this is also an isobaric process (constant pressure).

$$\therefore W_{B(1 \rightarrow 2)} = P_{\text{sat}@220^\circ\text{C}} (v_2 - v_1) \quad \leftarrow \text{Work done during isobaric process}$$

From A-4:

$$P_{\text{sat}@220^\circ\text{C}} = 2319.6 \text{ kPa}$$

$$\text{then: } W_{B(1 \rightarrow 2)} = 2319600 (0.0691132 - 0.00119)$$

$$W_{B(1 \rightarrow 2)} = 157554.6547 \text{ J/kg} \quad \leftarrow \text{Still on a specific basis}$$

$$\text{then: } W_{B(1 \rightarrow 2)} = m W_{B(1 \rightarrow 2)} \quad \leftarrow \text{Total work done}$$

$$W_{B(1 \rightarrow 2)} = 1680.6723 \times 157554.6547$$

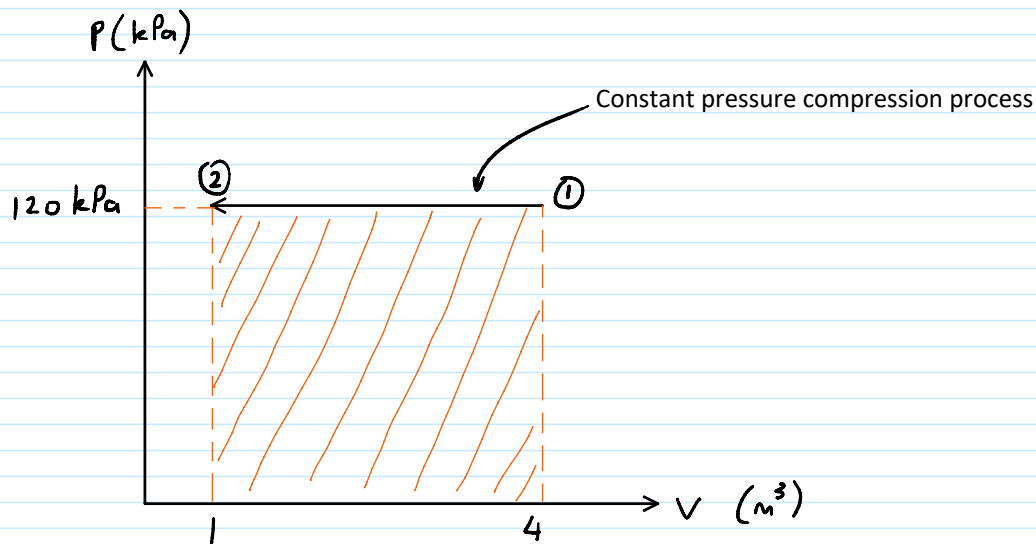
$$W_{B(1 \rightarrow 2)} = 264797.7439 \text{ kJ} \quad \leftarrow \text{A positive value for work indicates that work is done BY the water}$$

3. The volume of 1 kg of helium in a piston-cylinder device is initially 4 m³. Now the helium is compressed to 1 m³ while its pressure is maintained constant at 120 kPa. Determine the initial and final temperatures of helium as well as the work required to compress it, in kJ.

Knowns

- The working substance is helium, which means we can use the ideal gas assumption ($pV=mRT$)

Variable	State 1	State 2	Note
Mass	1 kg	1 kg	Closed system
Volume	4 m ³	1 m ³	Compression process (work is done on the helium)
Pressure	120 kPa	120 kPa	Isobaric process (constant pressure)



Work done

$$W_0 (1 \rightarrow 2) = P(V_2 - V_1) \text{ — Work done over constant pressure process}$$

$$W_0 (1 \rightarrow 2) = 120000 (1 - 4)$$

$$W_0 (1 \rightarrow 2) = -360000 \text{ J, or } -360 \text{ kJ} \text{ — The negative sign indicates that work was done ON the helium by the piston}$$

Initial and final temperature

$$PV = mR_{He}T \text{ — Ideal gas equation}$$

$$T = \frac{PV}{mR_{He}} \text{ — Rearranged to solve for temperature}$$

$$\text{From table A1, } R_{He} = 2.0769 \text{ kJ/kg} \cdot \text{K}$$

TABLE A-1

Molar mass, gas constant, and critical-point properties

Substance	Formula	Molar mass, M kg/kmol	Gas constant, R kJ/kg·K*	Critical-point properties		
				Temperature, K	Pressure, MPa	Volume, m ³ /kmol
Air	—	28.97	0.2870	132.5	3.77	0.0883
Ammonia	NH ₃	17.03	0.4882	405.5	11.28	0.0724
Argon	Ar	39.948	0.2081	151	4.86	0.0749
Benzene	C ₆ H ₆	78.115	0.1064	562	4.92	0.2603
Bromine	Br ₂	159.808	0.0520	584	10.34	0.1355
n-Butane	C ₄ H ₁₀	58.124	0.1430	425.2	3.80	0.2547
Carbon dioxide	CO ₂	44.01	0.1889	304.2	7.39	0.0943
Carbon monoxide	CO	28.011	0.2968	133	3.50	0.0930
Carbon tetrachloride	CCl ₄	153.82	0.05405	556.4	4.56	0.2759
Chlorine	Cl ₂	70.906	0.1173	417	7.71	0.1242
Chloroform	CHCl ₃	119.38	0.06964	536.6	5.47	0.2403
Dichlorodifluoromethane (R-12)	CCl ₂ F ₂	120.91	0.06876	384.7	4.01	0.2179
Dichlorofluoromethane (R-21)	CHCl ₂ F	102.92	0.08078	451.7	5.17	0.1973
Ethane	C ₂ H ₆	30.070	0.2765	305.5	4.48	0.1480
Ethyl alcohol	C ₂ H ₅ OH	46.07	0.1805	516	6.38	0.1673
Ethylene	C ₂ H ₄	28.054	0.2964	282.4	5.12	0.1242
Helium	He	4.003	2.0769	5.3	0.23	0.0578
n-Hexane	C ₆ H ₁₄	86.179	0.09647	507.9	3.03	0.3677
Hydrogen (normal)	H ₂	2.016	4.1240	33.3	1.30	0.0649
Krypton	Kr	83.80	0.09921	209.4	5.50	0.0924
Methane	CH ₄	16.043	0.5182	191.1	4.64	0.0993
Methyl alcohol	CH ₃ OH	32.042	0.2595	513.2	7.95	0.1180
Methyl chloride	CH ₃ Cl	50.488	0.1647	416.3	6.68	0.1430
Neon	Ne	20.183	0.4119	44.5	2.73	0.0417
Nitrogen	N ₂	28.013	0.2968	126.2	3.39	0.0899
Nitrous oxide	N ₂ O	44.013	0.1889	309.7	7.27	0.0961
Oxygen	O ₂	31.999	0.2598	154.8	5.08	0.0780
Propane	C ₃ H ₈	44.097	0.1885	370	4.26	0.1998
Propylene	C ₃ H ₆	42.081	0.1976	365	4.62	0.1810
Sulfur dioxide	SO ₂	64.063	0.1298	430.7	7.88	0.1217
Tetrafluoroethane (R-134a)	CF ₃ CH ₂ F	102.03	0.08149	374.2	4.059	0.1993
Trichlorofluoromethane (R-11)	CCl ₃ F	137.37	0.06052	471.2	4.38	0.2478
Water	H ₂ O	18.015	0.4615	647.1	22.06	0.0560
Xenon	Xe	131.30	0.06332	289.8	5.88	0.1186

Initial	Final
$T_1 = \frac{p_1 V_1}{m R_{He}}$	$T_2 = \frac{p_2 V_2}{m R_{He}}$
$T_1 = \frac{120000 \times 4}{1 \times 2076.9}$	$T_2 = \frac{120000 \times 1}{1 \times 2076.9}$
$T_1 = 231.1137 \text{ K}$	$T_2 = 57.7784 \text{ K}$

4. A gas is compressed from an initial volume of 0.2 m³ to a final volume of 0.1 m³. During the quasi-equilibrium process, the pressure changes with volume according to the relation $P = aV + b$, where $a = -1500 \text{ kPa/m}^3$ and $b = 400 \text{ kPa}$. Calculate the work done during this process: (a) by plotting the process on a P-V diagram and finding the area under the process curve and (b) by performing the necessary integrations.

Knowns

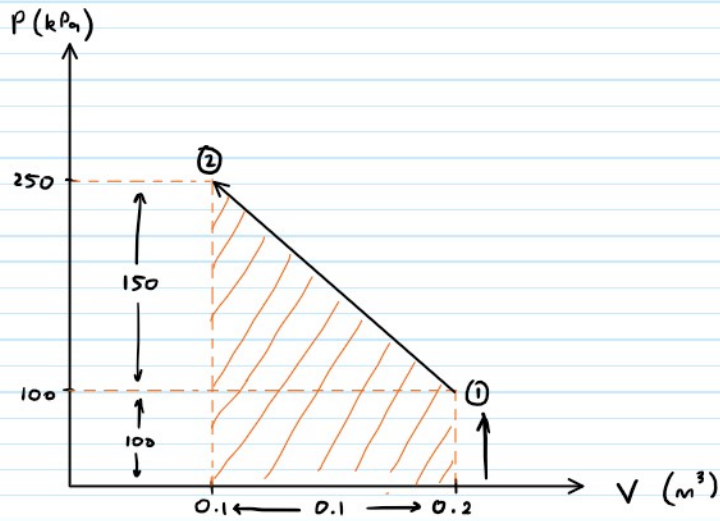
- The working substance is a gas, so the ideal gas assumption can be applied ($pV = mRT$)
- This is a quasi-equilibrium process, which means that it occurs extremely,

slowly, and so the system is always in a state of equilibrium. This means that the system's pressure and volume is always well-defined and uniform.

Variable	State 1	State 2	Note
Volume	0.2 m ³	0.1 m ³	Compression process

$$P = -1500V + 400 \leftarrow \text{Relationship between pressure and volume}$$

Initial pressure	Final pressure
$P_1 = -1500V_1 + 400$	$P_2 = -1500V_2 + 400$
$P_1 = -1500(0.2) + 1000$	$P_2 = -1500(0.1) + 400$
$P_1 = 100 \text{ kPa}$	$P_2 = 250 \text{ kPa}$



Graphical method

$$W_{B(1 \rightarrow 2)} = A_{\text{total}}$$

$$W_{B(1 \rightarrow 2)} = A_{\text{triangle}} + A_{\text{rectangle}}$$

$$W_{B(1 \rightarrow 2)} = \frac{(P_2 - P_1)(V_2 - V_1)}{2} + (100 - 0)(V_2 - V_1)$$

$$W_{B(1 \rightarrow 2)} = \frac{(250 - 100)(0.1 - 0.2)}{2} + (100 - 0)(0.1 - 0.2)$$

$$W_{B(1 \rightarrow 2)} = -17.5 \text{ J}$$

$\leftarrow$ This value is negative as it indicates that work is done ON the gas by the piston

Integral method

$$W_{B(1 \rightarrow 2)} = \int_{V_1}^{V_2} P dV$$

$$W_B(1 \rightarrow 2) = \int_{V_1}^{V_2} (-1500V + 400) dV$$

$$W_B(1 \rightarrow 2) = -1500 \int_{V_1}^{V_2} V dV + \int_{V_1}^{V_2} 400 dV$$

$$W_B(1 \rightarrow 2) = \left. -\frac{1500V^2}{2} \right|_{V_1}^{V_2} + 400V \Big|_{V_1}^{V_2}$$

$$W_B(1 \rightarrow 2) = -750(V_2^2 - V_1^2) + 400(V_2 - V_1)$$

$$W_B(1 \rightarrow 2) = -750(0.1^2 - 0.2^2) + 400(0.1 - 0.2)$$

$$W_B(1 \rightarrow 2) = -17.5 \text{ J}$$